

VETERINARY IMMUNOLOGY AND SEROLOGY • PART 1

Foundations, Immune System & Cells of Immunity

Covers the history and theories of immunology, primary and secondary lymphoid organs, cells of the innate and specific immune systems, innate and adaptive immunity, the complement system, and antigens including blood group and histocompatibility antigens.

Subjects

1

Parts

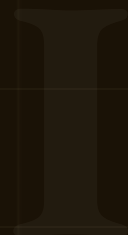
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Chapters

13

Edition

Extendable



HISTORY, LYMPHOID ORGANS, INNATE & ADAPTIVE IMMUNITY, COMPLEMENT, ANTIGENS (TABLE OF CONTENTS)

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◆ PART 1

VETERINARY IMMUNOLOGY AND SEROLOGY

History, Lymphoid Organs, Innate & Adaptive Immunity, Complement, Antigens

Systematic Veterinary Immunology and Serology (vetbvsc.in course notes)

History & Foundations of Immunology

IMMUNOLOGY

Branch of science dealing with the body's resistance to infection, or altered reactivity following exposure to a foreign substance. **Immunity** is the body's learned ability to resist infection; the **immune system** is the cells and molecules responsible; **immune response** is the reaction to a foreign antigen producing antibodies and lymphokines.

ANCIENT TO EARLY MODERN ERA **EXAM HOT**

- **430 BC** Thucydides (Athens) first mentioned immunity to a plague-like infection.
- **12th century** Chinese practised variolation — inhaling smallpox scab powder; deaths fell from 20% to 1%.
- **18th century** Rinderpest control in Europe by inserting nasal-discharge-soaked string into a dewlap incision.
- **1774** Benjamin Jesty inoculated his wife with vaccinia virus against smallpox.
- **1798** Edward Jenner inoculated an 8-year-old boy with cowpox blister fluid, protecting him from smallpox — termed **vaccination** (*vacca*, Latin for cow). Jenner is the Father of Immunology.

PASTEUR ERA

- **1879** Louis Pasteur discovered attenuated-culture immunity in chickens (*Pasteurella multocida*, fowl cholera) after an accidentally aged culture failed to kill.
- **1881** Pasteur publicly vaccinated sheep with heat-attenuated *Bacillus anthracis*, then challenged with virulent culture.
- **1885** Pasteur administered the first rabies vaccine to Joseph Meister (6 July 1885 — observed as Zoonosis Day); established the Pasteur Institute, Paris.
- **1885** Salmon (USA) showed heat-killed *Bacillus suispestifer* (now *Salmonella choleraesuis*) protects pigeons — first killed-vaccine demonstration.

- **1888** Roux & Yersin demonstrated bacterial toxins in diphtheria culture filtrate.
- **1890** Von Behring & Kitasato demonstrated toxin-antitoxin neutralisation and passive antibody transfer — Von Behring won the first Nobel Prize in Medicine, 1901.

CHRONOLOGY OF KEY ACHIEVEMENTS

EXAM HOT

YEAR	SCIENTIST(S)	FINDING
1862	Ernst Haeckel	Phagocytosis
1877	Paul Ehrlich	Mast cells
1891	Robert Koch	Delayed-type hypersensitivity
1894	Bordet, Pfeiffer & Isaeff	Bacteriolysis; complement
1900	Paul Ehrlich	Side chain (antibody formation) theory
1901	Karl Landsteiner	ABO blood group
1901	Bordet & Gengou	Complement Fixation Test
1902	Richet & Portier	Anaphylaxis
1903	Nicholas Arthus	Arthus phenomenon
1903	Almoth Wright	Opsonization
1906	Clemens Pirquet	Coined the term "allergy"
1912	Calmette & Guérin	BCG vaccine, tuberculosis
1917	Karl Landsteiner	Hapten
1934	John Marrack	Lattice theory of Ag-Ab binding
1936	Gorer	Major histocompatibility complex
1939	Tiselius & Kabat	Antibodies are gamma globulins
1944	Medawar & Burnet	Acquired immunological tolerance
1955	Glick	Bursa of Fabricius, B-cell site

YEAR	SCIENTIST(S)	FINDING
1957	Isaacs & Lindenmann	Interferon
1955-59	Burnet & Jerne	Clonal selection theory
1961-62	Miller & Good	Thymus role in cellular immunity
1975	Köhler & Milstein	Monoclonal antibodies
1976	Tonegawa	Gene rearrangement in antibody production
1995	Sakaguchi	Regulatory T cells



CHAPTER RECAP

- **Jenner, 1798** — smallpox vaccination using cowpox; Father of Immunology.
- **Pasteur, 1879-85** — attenuated vaccines for fowl cholera, anthrax, rabies.
- **6 July 1885** — first human rabies vaccination (Meister); Zoonosis Day.
- **Salmon, 1885** — first killed-vaccine demonstration (hog cholera bacillus in pigeons).
- **Von Behring & Kitasato, 1890** — antitoxin/passive transfer; Nobel Prize 1901.
- **Landsteiner, 1901** — discovered ABO blood groups.
- **Köhler & Milstein, 1975** — monoclonal antibody (hybridoma) technology.

Theories of Immunity — Cellular vs Humoral

Two schools of thought explained Pasteur's successful vaccination results before the mechanism of immunity was understood, developing side by side in the late 19th century.

CELLULAR IMMUNITY THEORY **EXAM HOT**

- **1882** Elie Metchnikoff (Russian zoologist) observed motile cells surrounding a rose thorn in transparent starfish larvae.
- **1883** Observed fungal spores attacked by blood cells in *Daphnia*; extended to mammalian leukocytes engulfing microbes, terming this **phagocytosis**.

Metchnikoff proposed phagocytosis as the main infection defence, mediated by two circulating phagocyte types: **microphages** (polymorphonuclear leukocytes) and **macrophages** (larger cells). Some pathologists criticised the theory, citing immunity observed without cells; it regained study focus ~50 years later.

HUMORAL THEORY

- **1886** Fodor observed direct anthrax-bacillus killing action of immune serum.
- **1888** George Nuttall observed bactericidal blood action in certain animals.
- **1889** Hans Buchner showed cell-free serum is bactericidal, but loses effect after 55 °C/1 h heating; the heat-labile substance was termed **alexin** (later cytolyisin, then **complement**).
- **1890** Von Behring & Kitasato demonstrated neutralising antitoxic serum activity against diphtheria/tetanus toxin — the first proof of humoral immunity.

CELLULAR THEORY (METCHNIKOFF)

- Phagocytosis is the primary defence
- Microphages (PMNs) and macrophages engulf microbes
- Criticised: immunity seen without cells present

HUMORAL THEORY

- Soluble blood/serum factors (antibody, alexin) mediate immunity
- Serum bactericidal action lost on heating (55 °C/1 h)
- Proven by antitoxin neutralisation and passive transfer

KEY POINT

Both theories are correct and complementary — cellular (phagocytic) and humoral (antibody/complement-mediated) mechanisms operate together in innate and adaptive immunity; this dual framework underlies the modern classification of immune components.



CHAPTER RECAP

- **Metchnikoff, 1882-83** — starfish larva/Daphnia observations; coined "phagocytosis".
- **Two phagocyte types** — microphages (PMNs) and macrophages.
- **Buchner, 1889** — named the heat-labile bactericidal serum factor "alexin" (= complement).
- **Heat lability test** — 55 °C for 1 hour destroys complement activity.
- **Von Behring & Kitasato, 1890** — first proof of humoral immunity via antitoxin transfer.
- **Both theories developed simultaneously** and are now understood as complementary.

Thymus

THYMUS

The first lymphoid organ to develop in mammals; a bilobed lympho-epithelial organ in the anterior mediastinum, developing from ectodermal invaginations (3rd bronchial cleft in humans).

LOCATION & GROWTH PATTERN **EXAM HOT**

In horse, cattle, sheep, pig and chicken it extends up the neck to the thyroid gland. Size increases through fetal and neonatal life, peaks at puberty, then atrophies (involutates) though remnants persist into old age.

STRUCTURE

Each lobe = lobules of loosely packed epithelial cells within a connective-tissue capsule, divided into outer **cortex** and inner **medulla**.

CORTEX

- Thymocytes (developing T cells) at various stages, reticular epithelial cells (stellate, desmosome-linked), few macrophages
- Capillaries have thick basement membrane + continuous epithelial layer — the **blood-thymus barrier** excluding antigen
- Immature T-lineage cells (yolk sac/bone marrow/fetal liver derived) enter via blood vessels and mature while migrating toward medulla

MEDULLA

- Few epithelial (reticular) cells, occasional **Hassall's corpuscles** (whorled epithelial-cell remnants)
- Mostly mature T cells bearing CD4+ or CD8+ markers
- Only mature medullary T cells exit via efferent vessels to blood/peripheral lymphoid tissue

SELECTION & HORMONES

EXAM HOT

During cortex-to-medulla migration, thymocytes contact epithelial cells, macrophages and dendritic cells, undergoing **positive selection** (imposes self-MHC restriction) and **negative selection** (central tolerance, removes self-reactive clones).

90-95%

thymocytes die by apoptosis in the thymus

5-10%

mature to CD4+ or CD8+ T cells

Thymic epithelial cells secrete hormones — **thymopoietins, thymosins, thymulin, thymostimulins** — driving maturation.

FUNCTIONS

- Differentiation of immature thymocytes into immunocompetent T lymphocytes.
- Colonisation of secondary lymphoid organs (spleen, lymph nodes) by thymus-derived lymphocytes.



CHAPTER RECAP

- **First lymphoid organ** to develop; bilobed, ectodermal origin.
- **Blood-thymus barrier** — thick basement membrane + epithelial layer excludes antigen from cortex.
- **Positive selection** = self-MHC restriction; **negative selection** = central tolerance.
- **90-95% apoptosis rate** — only 5-10% of thymocytes mature and exit.
- **Hassall's corpuscles** — whorled epithelial remnants unique to medulla.
- **Involutes after puberty** but remnants persist lifelong.

Other Primary Lymphoid Organs

PRIMARY LYMPHOID ORGANS

Also called generative/central lymphoid organs — where lymphocytes differentiate and mature into immunocompetent cells without antigenic stimulation. Includes bone marrow, thymus, bursa of Fabricius, some Peyer's patches and fetal liver.

BURSA OF FABRICIUS

EXAM HOT

Found only in birds (mammalian GALT equivalents: Peyer's patches, colon/appendix lymphoid tissue, tonsil). A round (elongated in ducks) sac from the dorsal cloacal epithelial diverticulum, above the cloaca. Reaches maximum size by 3 weeks post-hatch, then involutes.

- Structure: epithelial-lined sac connected to cloaca by a duct; folds (plicae) contain bursal follicles with a cortex (lymphocytes, plasma cells, macrophages) and medulla.
- Bursal microenvironment hormone: **bursepoietin/bursin** drives stem-cell differentiation into B cells.
- B cells appear in bursa 12-15 days into embryonic development.
- Not a pure primary organ — traps antigen and does some antibody synthesis; contains a small T-cell focus near the duct opening.

PEYER'S PATCHES

EXAM HOT

Some ruminant intestinal PP function as primary lymphoid organs. Sheep have two types:

TYPE	LOCATION	PERSISTENCE	CELLS
Jejunal PP	Jejunum	Throughout life	Pear-shaped follicles, up to 30% T cells
Ileo-caecal PP	Ileum/caecum	Max size before birth, gone by 15 months	Resembles avian bursa; B-cell maturation, negative selection

BONE MARROW

Primary lymphoid organ for B-cell development except in ruminants/pigs (ileo-caecal PP). Site of all circulating blood cell generation in adults, including B-cell maturation. Embryonic haematopoiesis: yolk-sac blood islands + para-aortic mesenchyme → liver/spleen → bone marrow after birth. At puberty, occurs mainly in flat bones (sternum, vertebrae, iliac bone, ribs).

- Red marrow: sponge-like reticular framework lined with endothelium between long trabeculae.
- Spaces hold fat cells, stem cells, haematopoietic cells; precursors exit via vascular sinuses.
- All blood cells except T lymphocytes are produced here.

LYMPHOEPITHELIAL GLANDS

Found throughout the large intestine/caecum of horses, dogs, pigs and other mammals — lymphoid aggregates surrounding elongated intestinal/caecal glands forming submucosal diverticula, resembling avian bursa. Rich in plasma cells, suggesting a major antibody-production site.



CHAPTER RECAP

- **Bursa of Fabricius** — bird-only; site of B-cell maturation; bursin hormone drives differentiation.
- **Ileo-cecal PP** — resembles avian bursa; disappears by 15 months in sheep.
- **Jejunal PP** — persists lifelong, up to 30% T cells.
- **Bone marrow** — B-cell primary organ except in ruminants/pigs (use ileo-cecal PP instead).
- **All blood cells except T cells** arise from bone marrow.
- **Lymphoepithelial glands** — large-intestine structures resembling bursa in horses, dogs, pigs.

Lymph Nodes & Lymphocyte Circulation

LYMPH NODES

Bean-shaped structures along lymphatic vessels, trapping antigen carried by lymph; covered by a fibrous capsule with outer cortex, para-cortical zone, and inner medulla, penetrated by lymphatic sinuses.

STRUCTURE

EXAM HOT

CORTEX — B-cell zone

- Primary follicles (naïve mature B cells)
- Secondary follicles with **germinal centres** — sites of B-cell somatic maturation, Ig class switching, memory-cell formation
- Follicular dendritic cells (FDCs) trap antigen, present to B cells
- ~70% of cortical T cells are CD4+ (helper)

PARACORTEX

- T-cell homing zone; interdigitating dendritic cells capture/process/present antigen to T cells

MEDULLA

- B cells, macrophages, reticular cells, plasma cells in cellular cords between sinuses
- Activated T cells exit to circulation; activated B cells migrate to germinal centre/medulla and secrete antibody



SPECIES VARIATION

Pigs, elephants, hippopotamuses, rhinoceroses and dolphins have lymph nodes made of aggregated nodules with cortex centrally and medulla peripherally — the reverse of the usual arrangement.

HEMOLYMPH NODES & DENDRITIC CELLS

Hemolymph nodes resemble lymph nodes in ruminants and other mammals; function unclear. **Dendritic cells** occur in skin, respiratory tract, GI tract and most parenchymal organs epithelia; lobulated nuclei, clear cytoplasm with **Birbeck granules**; capture and transport antigen to peripheral lymphoid organs; all derive from bone marrow (lymphoid or myeloid precursor lineage), related to mononuclear phagocytes.

LYMPHOCYTE CIRCULATION **EXAM HOT**

Blood lymphocytes are predominantly T cells, leaving blood by two routes:

- Naïve T cells bind **high endothelial venules (HEV)** in the paracortex — tall rounded endothelial cells with discontinuous junctions (absent in sheep lymph nodes).
- Memory T cells leave via conventional blood vessels in tissues, reaching nodes via afferent lymph.

90%

of node-leaving lymphocytes entered via HEV

10%

entered via afferent lymph

Lymph leaves via efferent lymphatics → large vessels → **thoracic duct** (largest lymph vessel, collects from lower body/intestine) → anterior vena cava. Thoracic duct cannulation drops blood/paracortical T cells sharply, confirming recirculation. In pigs, circulating lymphocytes enter nodes via HEV but leave the same way (not via lymphatics), so pig lymph has very few lymphocytes.



CHAPTER RECAP

- **Germinal centres** — B-cell somatic maturation, class switching, memory formation.
- **Paracortex** — T-cell zone, interdigitating dendritic cells.
- **Reversed cortex/medulla** — seen in pigs, elephants, hippos, rhinos, dolphins.
- **Birbeck granules** — characteristic dendritic-cell cytoplasmic granules.
- **HEV** — route for 90% of node-leaving lymphocytes; absent in sheep nodes.
- **Thoracic duct** — largest lymph vessel; cannulation confirms T-cell recirculation.
- **Pig lymph** — nearly lymphocyte-free (both entry and exit via HEV).

Spleen & Other Secondary Lymphoid Tissues

SPLEEN

Major site of immune response to blood-borne antigens (lymph nodes filter lymph; spleen filters blood). Removes antigen particles and aged blood cells; stores red cells/platelets; produces blood cells in the fetus.

STRUCTURE **EXAM HOT**

Covered by capsule with trabeculae extending into splenic pulp, divided into red and white pulp, supported by a reticular-cell/fibre meshwork. Single splenic artery pierces the capsule at the hilum, dividing into trabecular arteries.

WHITE PULP

- Periarterial lymphoid sheath (**PALS**) — T-cell cuff around trabecular (central) arterioles
- B-cell follicles within PALS, some with germinal centres (**Malpighian corpuscles**)
- Mantle zone — T-cell layer surrounding each follicle

MARGINAL ZONE

- Separates white and red pulp; fine reticular meshwork; contains both T and B cells

RED PULP

- Cords of reticular meshwork + sinuses; contains RBCs, macrophages, platelets, plasma cells
- Filters RBCs/WBCs; site of extramedullary haematopoiesis
- Destroys platelets/RBCs — termed **hemocatharesis**

FUNCTIONS

- Site for development of immune response (occurs in white pulp).

- Site for haematopoiesis (lymphopoiesis and erythropoiesis).
-
- Organ for filtration of RBCs and WBCs.
-
- Traps antigen and mounts an immune response.

CUTANEOUS IMMUNE SYSTEM

Epidermis: keratinocytes, melanocytes (both innate-immunity effectors), epidermal **Langerhans cells**, intraepithelial T cells (~2% of epidermal cells, majority CD8+). Langerhans cells capture antigen, migrate (chemokine-driven) to dermis then to lymph nodes via lymphatics. Dermis: CD8+ and CD4+ T cells, mainly perivascular, with scattered macrophages.

MUCOSAL IMMUNE SYSTEM (MALT)

Defends GI, respiratory, genital tract and oral cavity mucosal surfaces. In the GI tract, lymphocytes occur in three regions: intraepithelial (mainly CD8+ T cells), lamina propria (scattered), and Peyer's patches of lamina propria. Follicles hold B cells (IgM at periphery, IgA in germinal centre), macrophages, dendritic cells; T cells predominate in inter-/para-follicular regions.

BONE MARROW AS A SECONDARY ORGAN

Probably the largest secondary lymphoid tissue. IV antigen is partly trapped here. During primary response, antibodies form largely in spleen/lymph node; late in the response, memory cells colonise bone marrow. On second antigen dose, bone marrow produces very large antibody quantities — the major source of IgG in adult rodents.



CHAPTER RECAP

- **PALS** — T-cell sheath around the central (trabecular) artery in white pulp.
- **Marginal zone** — separates red/white pulp, mixed T+B cells.
- **Hemocatharesis** — red-pulp destruction of aged RBCs/platelets.
- **Langerhans cells** — epidermal antigen-capturing DCs; migrate to lymph nodes.
- **GI lymphocyte sites** — intraepithelial, lamina propria, Peyer's patches.
- **Bone marrow (secondary role)** — major IgG source after memory-cell colonisation post-primary response.

Cells of the Non-Specific Immune System

INNATE IMMUNE CELLS (MYELOID LINEAGE)

Non-specific immune response is the first line of defence; phagocytes (neutrophils and macrophages) trap and destroy invading microorganisms.

NEUTROPHILS

EXAM HOT

First line of defence — "martyrs of the immune system"; recruited by chemotactic signals; macrophages follow 5-6 hours later. Major myeloid cell type; originate from bone marrow, ~2 weeks development; acquire cytoplasmic granules during myelocyte stage (**polymorphonuclear, PMN**). Circulate 10-12 h then move into tissues; total lifespan a few days; do not multiply. ~12 µm diameter, segmented nucleus.

60-75%

of blood leucocytes in carnivores

20-30%

in ruminants

50%

in horses

GRANULE TYPE	CONTENTS	FUNCTION
Primary (azurophilic)	Defensins, myeloperoxidase, lysozyme, elastase, β-glucuronidase, cathepsin B	Bactericidal; respiratory burst; degrade bacterial products; destroy cell wall
Secondary (specific)	Lysozyme, collagenase, lactoferrin, gelatinase	Destroy cell wall; bind iron; degrade connective tissue

Species nuclear morphology: canine/feline neutrophils lack true filament formation between lobes; bovine has filaments (though many identified only by nuclear indentation); equine shows dark-staining chromatin plaques giving a granular nucleus appearance. Neutrophils destroy ingested material but cannot process antigen for presentation.

MACROPHAGES

Immature forms produced in bone marrow, released as **monocytes** (~5% blood leukocytes); mature into macrophages after tissue migration. Rarely divide; radiation-resistant; lifespan 3-4 months. 15-20 µm, bean-

shaped/round nucleus, rich in mitochondria/lysosomes/RER/Golgi. Differentiate into: microglia (CNS), Kupffer cells (liver), alveolar macrophages (lung), osteoclasts (bone), splenic/peritoneal macrophages, dendritic cells (lymphoid organs), Langerhans cells (skin), mesangial cells (kidney).

- Phagocytosis of foreign particles.
- Antigen processing/presentation for the immune response.
- Secretion of >100 bioactive molecules: enzymes (lysozyme, collagenase, proteases, elastases), immune mediators (IL-1, interferon- α , complement components), inflammatory mediators (prostaglandins, thromboplastins, plasminogen activators).
- Immune regulation — correct antigen presentation, dose regulation, monokine secretion.
- Antitumour activity — TNF- α , hydrolytic enzymes, H₂O₂, superoxide, IL-1.
- Wound healing — proteases clear damaged tissue; growth factors stimulate fibroblast collagen secretion and angiogenesis.

| EOSINOPHILS

Second major granulocyte, named for intense eosin staining. Bone marrow origin, mature in spleen; short lifespan — blood half-life ~30 min, tissue half-life ~12 days; tissue eosinophils ~500× blood count. Slightly larger than neutrophils, bilobed nucleus.

- Primary granules — arylsulfatase, peroxidase, acid phosphatase.
- Crystalloid granules — core of major basic protein + matrix (eosinophil cationic protein, eosinophil peroxidase, eosinophil-derived neurotoxin).
- Function: extracellular parasite killing via **antibody-dependent cellular cytotoxicity (ADCC)** using Fc receptors specific for IgE; helminths resist neutrophil/macrophage lysosomal enzymes but not eosinophil major basic/cationic protein.
- Participate in Type I hypersensitivity (IgE-mediated) tissue damage.

| BASOPHILS & MAST CELLS

Basophils — bone marrow derived, smallest granulocyte, IgE Fc receptors, histamine-rich granules; not normally in extravascular tissue but may infiltrate under lymphocyte influence. **Mast cells** — bone marrow derived, resident cells essential for IgE-mediated inflammation; granules rich in histamine, heparin, TNF- α and

other mediators, plus superoxide dismutase/peroxidase/acid hydrolases; scattered in connective tissue beneath skin, lung alveoli, GI/nasal mucosa.



CHAPTER RECAP

- **Neutrophil development** — ~2 weeks in bone marrow; circulate only 10-12 h.
- **Primary vs secondary granules** — azurophilic (defensins, MPO) vs specific (lysozyme, lactoferrin, collagenase).
- **Neutrophils cannot present antigen** — unlike macrophages.
- **Macrophage differentiation** — microglia, Kupffer cells, alveolar macrophages, osteoclasts etc.
- **Eosinophils** — ADCC via IgE-Fc receptor, kill helminths with major basic protein.
- **Mast cells vs basophils** — both IgE-bearing, histamine-rich; mast cells are tissue-resident, basophils circulate.

Lymphocytes — B Cells & T Cells

LYMPHOCYTES

Cells recognising foreign antigen and mounting immune response; small round cells (7-15 μm) with a large hematoxylin-staining nucleus and thin cytoplasmic rim. Originate from bone marrow stem cells, mature/differentiate into B and T cells in lymphoid organs; highly mobile, circulating blood $\rightleftharpoons$ tissue $\rightleftharpoons$ lymph.

B LYMPHOCYTES

EXAM HOT

Origin: bone marrow multipotent stem cells; maturation in fetal liver (before birth), then bone marrow or Peyer's patches (mammals) or bursa (birds). Principal maturation event: Ig gene rearrangement/expression. Mature B cells are not antibody-secreting until antigenic stimulation differentiates them into **plasma cells** — large (2 $\times$ lymphocyte size), eccentric nucleus, abundant RER, non-dividing, 2-3 day lifespan, secreting ~ 300 Ig molecules/second. Non-plasma-cell B cells become **memory cells**, some surviving >20 years without antigen.

CLUSTER OF DIFFERENTIATION (CD) SYSTEM

Surface markers identified by monoclonal antibodies, uniformly named by "cluster" designation (CD1, CD2...); >200 CD molecules defined; incompletely characterised ones are "workshop" candidates (CDw).

- B-cell surface receptors: BCR (Ig + Ig α + Ig β), immunoglobulin receptors, complement receptors, cytokine receptors, MHC Class II, CD19, CD20, CD21, CD40.

T LYMPHOCYTES

Progenitors originate in bone marrow, are attracted to thymus at any embryonic/post-embryonic time. Immature thymocytes populate the cortex, undergo TCR gene rearrangement, express CD3/TCR/CD4/CD8, migrate to medulla for maturation.

- T-cell receptors (α/β and γ/δ) with CD3 complex.
- CD2 — receptor for CD58; CD4 — helper T-cell marker; CD8 — cytotoxic T-cell marker.

- CD71 — transferrin receptor; CD35 — complement receptor; CD25 — IL-2 receptor.
- Adherence molecules — integrin family, selectin family.

T CELL SUBSETS

EXAM HOT

SUBSET	MARKER	FUNCTION
Th1	CD4+	Cytokines inducing cell-mediated immunity
Th2	CD4+	Cytokines inducing humoral immunity
Th17	CD4+	Inflammation in autoimmune/inflammatory disease
Cytotoxic (Tc)	CD8+	Destroy virus-infected/tumour cells; transplant rejection
γ/δ T cells	Non-polymorphic TCR	Recognise microbial glycolipids via CD1+ APCs; ~60% of blood lymphocytes in young ruminants (5-15% in human/mice)
Memory T cells	CD4+ or CD8+	Persist long-term; rapid effector expansion on re-exposure
Regulatory T cells	—	Maintain peripheral tolerance/immunity balance
NKT cells	CD1-restricted	Bridge innate/adaptive immunity; recognise glycolipid antigens

B VS T CELL COMPARISON

EXAM HOT

FEATURE	B CELLS	T CELLS
Maturation site	Bursa/bone marrow/Peyer's patches	Thymus
Distribution	LN cortex, splenic follicles	LN paracortex, splenic PALS
Circulate	No	Yes
Life span	Days to weeks	6 months-10 years
Surface Ig	Present (IgM,IgD,IgG)	Absent

FEATURE	B CELLS	T CELLS
Immunity type	Humoral	Cell-mediated
Secreted product	Immunoglobulins	Cytokines
Mitogen response	Pokeweed, LPS	PHA, ConA, BCG, pokeweed
EAC rosette	Yes (CR2/CR1)	No
E/SRBC rosette	No	Yes (CD2)
Antigen uptake	Unprocessed antigen	Only processed antigen
Subsets	Very few	Many



CHAPTER RECAP

- **Plasma cells** — secrete ~300 Ig molecules/second, 2-3 day lifespan.
- **Memory B cells** — can persist >20 years without antigen.
- **γ/δ T cells** — ~60% of blood lymphocytes in young ruminants (unusually high).
- **CD4 vs CD8** — helper vs cytotoxic/suppressor T-cell markers.
- **EAC rosette** — B-cell test (complement receptor); **E-rosette** — T-cell test (CD2/sheep RBC).
- **Th1/Th2/Th17** — CMI-inducing / humoral-inducing / inflammatory subsets.
- **B cells only take up unprocessed antigen**; T cells require processed peptide.

Innate Immunity — Barriers, Phagocytosis & NK Cells

INNATE (NON-SPECIFIC/NATURAL/NATIVE) IMMUNITY

First line of defence, independent of prior exposure. Components: physical/chemical barriers, biological barriers (phagocytes, NK cells), humoral factors (complement, inflammatory mediators), cytokines, genetic factors.

PHYSICAL & CHEMICAL BARRIERS EXAM HOT

- **Skin** — keratinised layer impermeable to microbes; dry environment limits growth; sweat lactic acid/salts inhibit bacteria/fungi; normal flora inhibits superinfection; sebaceous sebum protects; epithelial peptides act as natural antibiotics.
- **Mucous membrane** — stratified squamous epithelium + mucus traps microbes; gastric HCl/enzymes kill microbes; intestinal **cryptocidins** (antimicrobial peptides); acidic vaginal pH kills microbes; tears contain lysozyme.

BIOLOGICAL BARRIERS — PHAGOCYTE FUNCTIONS EXAM HOT

01 RECRUITMENT

Chemotactic gradients (C5a, fibrinopeptide B, platelet factor 4) draw neutrophils/monocytes; endothelial adhesion molecules (triggered by LPS/cytokines) mediate rolling → diapedesis into tissue.

02 RECOGNITION

Mannose-binding receptors (bind terminal mannose/fructose on bacterial glycoproteins), opsonin receptors (opsonisation = coating for phagocytosis; **opsonic index** = ratio of opsonin in diseased vs healthy serum), Toll-like receptors (TLRs, pattern recognition of microbial molecules).

03 PHAGOCYTOSIS

Pseudopod extension (actin/myosin), opsonin-receptor binding, engulfment forming a **phagosome**.

04

DESTRUCTION — OXYGEN-DEPENDENT

Respiratory burst (reactive oxygen intermediates, ROIs) via NADPH oxidase → superoxide → H₂O₂ → myeloperoxidase converts to microbicidal hypohalides; nitric oxide (via inducible NOS) forms further reactive radicals.

05

DESTRUCTION — OXYGEN-INDEPENDENT

Primary granules fuse with phagosome (**phagolysosome**); proteases (elastase), lipases, nucleic-acid enzymes digest the microbe.

06

CYTOKINE PRODUCTION

Monokines (macrophage-derived) or lymphokines (lymphocyte-derived); **interleukins** numbered chronologically IL-1 to IL-35; mediate/regulate innate and adaptive immunity, stimulate haematopoiesis, recruit/activate leukocytes.

KEY POINT

Neutrophils circulate only ~6 h; undergo programmed cell death if not recruited within hours, cleared by liver/spleen resident macrophages. All mononuclear phagocytes (bone-marrow derived monocytes → tissue macrophages: histiocytes, Kupffer cells, microglia, alveolar/mesangial macrophages) form the **reticuloendothelial (RE) system**.

NATURAL KILLER (NK) CELLS**EXAM HOT**

5-20% of mononuclear cells in blood/spleen, rare elsewhere. Kill target cells without prior activation; neither T nor B lineage — no Ig or TCR expression. Large granular lymphocytes from bone marrow precursors.

- Bear activating and inhibitory receptors — inhibitory dominates when both engaged, protecting normal host cells.
- Infected cells often lose MHC Class I expression (viral immune evasion), removing inhibitory-receptor ligands → NK activation.
- Granules contain **perforin** (forms membrane pores) and **granzymes** (enter pores, trigger apoptosis).
- Kill virus-infected and intracellular bacteria-infected cells; IFN-γ from NK cells aids macrophage phagocytosis; participate in ADCC.

GENERAL BARRIERS (GENETIC & PHYSIOLOGICAL)

- Species specificity — rinderpest affects cattle/sheep/goats, not humans; FMD virus does not infect horses or dogs.
- Genetically resistant groups — e.g. African-descent individuals more TB-susceptible in America; white-line (B1) broiler chickens more susceptible to hydropericardium syndrome.
- Cold-blooded animals resist tetanus toxin.
- Avian body temperature (42 °C) restricts many pathogens; fever is beneficial.
- Age — very young and old are most susceptible to infection.



CHAPTER RECAP

- **Opsonic index** = ratio of opsonin, diseased vs healthy serum.
- **Respiratory burst** — NADPH oxidase → ROIs, myeloperoxidase converts H₂O₂ to hypohalides.
- **RE system** — all mononuclear phagocytes collectively (histiocytes, Kupffer, microglia etc).
- **NK cells** — neither T nor B; kill via perforin + granzyme; inhibited by self-MHC I.
- **Loss of MHC I** — the trigger that activates NK killing of infected cells.
- **Neutrophil circulation** — only ~6 hours before recruitment or apoptosis.
- **Species/genetic barriers** — rinderpest, FMD host restriction are classic examples.

Adaptive Immunity — Features & Types

ADAPTIVE (ACQUIRED/SPECIFIC) IMMUNITY

Specific for microbial/non-microbial antigens, increased by repeated exposure. B-lymphocyte mediated = **humoral immunity**; T-lymphocyte mediated = **cell-mediated immunity (CMI)**. Develops as a response and adapts to infection, unlike innate immunity.

INNATE VS ADAPTIVE FEATURES

EXAM HOT

FEATURE	INNATE	ADAPTIVE
Specificity	Shared microbial structures	Recognises particular antigen
Diversity	Limited, germ-line encoded	Very large — somatic gene recombination
Memory	No	Yes
Non-reactivity to self	Yes	Yes

TYPES & ACQUISITION

HUMORAL IMMUNITY — B lymphocytes, antibody neutralisation/elimination of microbes

CELLULAR IMMUNITY — T lymphocytes; effector cytokines activate macrophages; Tc cells kill infected cells directly

ACTIVE IMMUNITY

- Host produces antibody itself
- Develops slowly, persists long
- Natural (infection) or Artificial (vaccination)

PASSIVE IMMUNITY

- Antibody prepared elsewhere, introduced
- Rapid onset, short duration
- Natural (maternal/colostral transfer) or Artificial (immune serum, cell transfer)

CARDINAL FEATURES

EXAM HOT

- **Specificity** — response is highly specific for distinct antigens/epitopes; antigenic determinants (epitopes) confer this specificity.
- **Diversity** — total lymphocyte antigenic-specificity range = **lymphocyte repertoire**, discriminating 10^7 - 10^9 distinct determinants via variable receptor binding-site structure.
- **Memory** — sensitised lymphocytes respond more efficiently on re-exposure; secondary response has shorter lag, more antibody (IgG-dominant vs IgM-dominant primary response); memory T cells home to infection sites better than naïve T cells.
- **Specialization** — distinct, special responses to different microbes, maximising anti-microbial defence.
- **Self-limitation** — all responses wane with time (homeostasis) once antigen is eliminated; activated lymphocytes die or become quiescent memory cells.
- **Non-reactivity to self** — discriminates self vs non-self; failure → **autoimmune disorder**.

CELLULAR COMPONENTS & RESPONSE COMPONENTS

Lymphocytes, antigen-presenting cells (APCs — mostly dendritic cells) and effector cells (activated T lymphocytes, mononuclear phagocytes, other leukocytes) constitute the system. Components of the immune response: recognition of antigen → processing → production of antibody/cytokines → antigen-antibody reaction or CMI response → elimination of antigen (or succumbing to infection).

HUMORAL VS CELL-MEDIATED IMMUNITY

EXAM HOT

CHARACTERISTIC	HUMORAL	CELL-MEDIATED
Antigen	Extracellular	Intracellular
Responding cells	B lymphocytes	T lymphocytes
Effector function	Antibody-mediated elimination	Lymphokine-mediated elimination/lysis
Passive transfer	Through serum	Through T cells



CHAPTER RECAP

- **Lymphocyte repertoire** — discriminates 10^7 - 10^9 distinct antigenic determinants.
- **Memory response** — shorter lag, IgG-dominant (vs IgM-dominant primary).
- **Self-limitation = homeostasis** — response wanes once antigen is cleared.
- **Autoimmune disorder** — the failure mode of self/non-self discrimination.
- **Passive transfer test** — humoral via serum, CMI via T cells (not serum).
- **APCs** — mostly dendritic cells, capture and present antigen.

Complement System

COMPLEMENT SYSTEM

A major humoral/innate effector mechanism — at least 19 heat-labile (56 °C/30 min) serum proteins, ~10% of the globulin fraction of serum, molecular weight 24 kDa (factor D) to 460 kDa (C1q).

NOMENCLATURE & ACTIVATION EXAM HOT

Numeric prefix C (C1-C9) or alphabetic (B, D, P); peptide fragments = lowercase suffix (C3a, C3b); enzymatically active complexes get a bar over the symbol. Proteins requiring proteolytic activation are **zymogens**; sequential activation = **cascade reaction**. Guinea pig serum (fresh, pooled) is the best complement source; haemolysed or tissue-damaged blood releases anticomplement factors and must be avoided for CFT. Mouse and horse complement is incomplete/poorly lytic.

THREE ACTIVATION PATHWAYS

- **Classical pathway** — activated by antibody (IgM/IgG) bound to antigen
- **Alternative pathway** — activated by microbial cell surface, no antibody needed
- **Lectin pathway** — activated by plasma lectin bound to mannose residues

CLASSICAL PATHWAY EXAM HOT

01 C1 BINDING

C1 (C1q+C1r+C1s, Ca^{2+} -linked) binds the CH3 domain of IgM or CH2 of IgG; ≥ 2 IgG Fc regions needed (IgM pentamer activates with one molecule — making IgM the more efficient complement fixer).

02 C1 ACTIVATION

C1q binding to ≥ 2 Fc regions activates C1r, which cleaves/activates C1s (serine esterases); normally blocked by C1-INH, overcome once Ig binds antigen.

03 C4 CLEAVAGE

C1s cleaves C4 $\rightarrow$ C4a (released) + C4b (binds target surface).

04

C3 CONVERTASE FORMATION

C2 binds C4b (Mg^{2+} -dependent) $\rightarrow$ C4b2; C1s splits C2 into C2a+C2b (substrate modulation) $\rightarrow$

C4b2a = classical C3 convertase.

05

C3 CLEAVAGE

C4b2a cleaves C3 $\rightarrow$ C3a (released) + C3b (deposits on surface); can also feed the alternative pathway.

06

C5 CONVERTASE

C4b2a3b = classical C5 convertase, cleaves C5, initiating the terminal pathway.

ALTERNATIVE PATHWAY

C3 spontaneously breaks to C3a+C3b; C3b binds covalently to cell surfaces. On mammalian (sialylated) cells, H factor binds C3b, activates I factor (protease), destroying C3b — no activation. On microbial surfaces (bacterial walls, LPS, viruses, aggregated IgA, cobra venom), C3b instead binds factor B; factor D cleaves bound B $\rightarrow$ Bb (+soluble Ba); **C3bBb = alternative C3 convertase** (half-life 5 min). Properdin (factor P) stabilises it to C3bBbP (half-life 30 min). C3bBb3b = alternative C5 convertase.

LECTIN PATHWAY

Macrophage ingestion stimulates IL-1, IL-6, TNF- α $\rightarrow$ hepatocytes secrete acute-phase **mannose-binding protein (MBP/MBL)**, structurally similar to C1q, activating C1r-C1s (classical pathway) via MBP-associated serine esterase. Cattle/buffalo have additional mannose-binding proteins including **conglutinin**, which clumps C3b-coated particles.

TERMINAL PATHWAY

EXAM HOT

Any C5 convertase cleaves C5 $\rightarrow$ C5a (released) + C5b (binds C3b) $\rightarrow$ binds C6+C7 $\rightarrow$ C5b67 (detaches, inserts into nearby lipid bilayer) $\rightarrow$ binds C8 + 12-18 molecules of C9 $\rightarrow$ **membrane attack complex (MAC)**, C5b678(9)_n — a transmembrane doughnut-shaped pore causing osmotic lysis.

REGULATION

REGULATOR	ACTION
C1-INH	Serine protease inhibitor; dissociates C1r/C1s from C1q
Factor I	Serine protease; cleaves C3b/C4b (with H, MCP, C4BP, CR1 as cofactors)
Factor H	Binds C3b, displaces Bb; cofactor for Factor I on C3b

REGULATOR	ACTION
C4-binding protein	Binds C4b, displaces C2; cofactor for Factor I on C4b
MCP	Cofactor for Factor I cleavage of C3b/C4b
DAF	Displaces C2b from C4b and Bb from C3b
Vitronectin/clusterin/protectin	Block C9 binding, prevent MAC formation

FUNCTIONS

EXAM HOT

- Opsonisation & phagocytosis — C3b/C4b-coated microbes bind specific macrophage/neutrophil receptors.
- Complement-mediated cytotoxicity — via MAC/osmotic lysis.
- Removal of immune complexes.
- Activates B cells — signals initiation of humoral response.
- Anaphylaxis — C3a, C4a, C5a ("anaphylatoxins") trigger mast-cell degranulation/histamine release.



CHAPTER RECAP

- **≥19 proteins**, heat-labile at 56 °C/30 min, ~10% of serum globulin.
- **IgM > IgG** at complement fixation — one IgM molecule suffices; IgG needs two adjacent molecules.
- **C4b2a** = classical C3 convertase; **C3bBb** = alternative C3 convertase.
- **Properdin** extends alternative C3 convertase half-life from 5 to 30 minutes.
- **MAC = C5b678(9)n** — transmembrane pore causing osmotic lysis.
- **Anaphylatoxins** — C3a, C4a, C5a trigger mast-cell degranulation.
- **Guinea pig serum** — standard complement source for CFT.

Antigen — Properties & Antigenicity

ANTIGEN



A substance that, introduced into a susceptible animal's tissues, stimulates formation of specific antibody and/or lymphokines with which it reacts specifically and observably.

IMMUNOGENICITY VS ANTIGENICITY

EXAM HOT

- **Immunogenicity** — ability to induce a humoral and/or cell-mediated immune response; such material = **immunogen**.
- **Antigenicity** — ability of a molecule to be recognised by antibody or lymphocyte.
- All immunogens are antigenic, but not all antigens are immunogenic.

MOLECULAR SIZE

- Larger molecular weight → better antigen (e.g. ovalbumin, 40,000 Da)
- Exception: glucagon (2,600 Da) is a good antigen
- Penicillin/aspirin (<1,000 Da) are poor antigens

COMPLEXITY

- Complex proteins > repeating polymers (lipids, carbohydrates, nucleic acids)
- Bacterial exotoxins, egg albumin — powerful; haemoglobin — weak; protamines/histones/gelatin — almost devoid
- Complex carbohydrate bound to protein is more immunogenic (e.g. Gram-negative cell-wall antigens)

SHAPE

- Fixed shape → higher antibody response; highly flexible molecules are poor antigens

STABILITY

- Antigen must be stable/rigid — gelatin is poor unless cross-linked by tyrosine/tryptophan residues
- Flagellin is weak; stability/antigenicity enhanced by polymerisation

DEGRADABILITY

- Must be degradable to trigger response — stainless steel, plastic (inert polymers) are not antigenic

FOREIGNNESS

- Antigenicity $\propto$ degree of foreignness — same-species < related-species < unrelated-species antigens
- Self-antigens normally don't provoke antibody; exceptions = autoantibodies

GENOTYPE OF RECIPIENT

- Immune-response genetic control confined to MHC genes — MHC products present processed antigen to T cells

DOSAGE & ROUTE

- Insufficient dose fails to activate enough lymphocytes; excessive dose can induce non-responsive state; boosters needed for strong response; parenteral routes preferred

ADJUVANTS

- Enhance immunogenicity — prolong antigenic persistence, enhance co-stimulation, induce granuloma, non-specifically stimulate lymphocyte proliferation

ADJUVANT EXAMPLES

Alum, aluminium hydroxide, bacterial LPS, saponin, IL-12, Montanide, Freund's adjuvant. **Freund's incomplete adjuvant** = mineral oil + emulsifier (mannide monooleate); **Freund's complete adjuvant** = incomplete + heat-killed Mycobacteria — more potent.

HAPTEN **EXAM HOT**

HAPTEN

Usually non-protein, low-MW substance with little/no antigenicity alone, but antigenic when coupled to a carrier protein — a **partial antigen** (Greek *haptain*, "to fasten"). Cannot induce antibody formation alone but reacts specifically with existing antibody.

Test chemical (e.g. aniline, p-aminobenzoic acid) is diazotized and coupled to a protein forming a **conjugated antigen/azoprotein**. Complex hapten = high MW, polyvalent — forms visible precipitation with antibody against the complete antigenic complex. Simple hapten = low MW, univalent — no visible precipitation. Examples: penicillin, dinitrophenyl (DNP).

EPITOPES (ANTIGENIC DETERMINANTS) **EXAM HOT**

The smallest unit of antigenicity — specific chemical groupings/spatial configurations on the antigen surface, sensitising immunocytes and reacting with the complementary antibody/TCR site. The antibody-side combining region = **paratope/idiotope**; the T-cell recognition site = **agretope**. Most epitopes span 6-10 amino acids (T-cell epitopes) or 10-15 amino acids (B-cell epitopes).



CHAPTER RECAP

- **All immunogens are antigens; not all antigens are immunogens.**
- **MW <1000 Da** — penicillin, aspirin — usually poor antigens; glucagon (2600 Da) is an exception.
- **9 factors** govern antigenicity — size, complexity, shape, stability, degradability, foreignness, genotype, dose/route, adjuvants.
- **Freund's complete** = incomplete + heat-killed Mycobacteria.
- **Hapten** — antigenic only when carrier-conjugated; complex (visible precipitate) vs simple (no precipitate).
- **Paratope** — antibody combining site; **Agretope** — T-cell recognition site.
- **Epitope size** — 6-10 aa (T cell) vs 10-15 aa (B cell).

Types of Antigen, Blood Groups & MHC

CLASSIFICATION BY ANTIGENIC SPECIFICITY

EXAM HOT

SPECIES-SPECIFIC — specific to one host species

ORGAN-SPECIFIC — specific to one organ within a species (e.g. kidney protein differs from lung)

HETEROPHILE ANTIGENS — identical antigens shared across bacterial species and different animal tissues; antibody to one cross-reacts with the other

- **Forssman antigens** — glycoprotein in cell walls of Streptococcus, Shigella, Salmonella, Clostridium and on RBCs of horses, sheep, cats, mice; hosts bearing them don't make antibody (self-recognition)
- Antigen in Proteus strains OX-19, OX-2, OX-K and Rickettsial organisms — basis of the **Weil-Felix reaction**

ALLOANTIGENS (ISOANTIGENS) — antigenic to other individuals of the same species; corresponding antibodies = alloantibodies (e.g. blood group, transplantation antigens)

SYNGENIC ANTIGEN — present in individuals of identical genetic makeup (e.g. monozygotic twins)

AUTOANTIGEN — sequestered self-tissue that becomes antigenic if released into circulation (e.g. sperm, lens protein)

SUPERANTIGEN — activates T cells without APC processing (e.g. streptococcal A antigen)

BY T-CELL DEPENDENCY

T-DEPENDENT ANTIGEN

- Complex (RBC, serum proteins)
- Rapidly metabolised
- Requires T-cell help; produces memory + class switching

T-INDEPENDENT ANTIGEN

- LPS, pneumococcal polysaccharide, flagellin, lectins (Con A, PHA, pokeweed)
- Repeating-unit composition, slowly metabolised
- No memory or class switching

MICROBIAL ANTIGENS

- **Soluble** — toxins, enzymes excreted into environment.
- **Cellular** — **bacterial**: Somatic (O) antigen (LPS-protein, highly variable, common core has vaccine potential); Capsular (K) antigen (polysaccharide, e.g. E. coli K; or polypeptide, e.g. poly-D-glutamic acid in B. anthracis); Flagellar (H) antigen (flagellin protein); Fimbrial (F) antigen; Spore antigen (exosporidium); also porins, heat-shock proteins, exotoxins.
- **Viral** — structural VP (viral proteins), capsid protein, envelope (lipoprotein+glycoprotein), e.g. HN glycoprotein of Newcastle disease virus.
- **Non-microbial** — food substances, pollen grains, fungal spores, dust particles (allergic reactions).

CELL SURFACE ANTIGENS

Important types: blood group antigen, MHC molecules, cluster of differentiation (CD). Blood transfusion became scientifically feasible after **Landsteiner (1900)** discovered blood groups. Most blood group antigens are glycoproteins/glycolipids, integral cell-membrane components; do not participate in antigen processing but influence graft rejection.

ABO BLOOD GROUP SYSTEM **EXAM HOT**

4 groups by presence/absence of A and B antigens; group A subdivides into A1 (~80%) and A2 (~20%). Isoantibodies appear by ~6 months of age, persisting thereafter.

GROUP	RBC ANTIGEN	SERUM ANTIBODY	DONOR/RECIPIENT ROLE
A	A	Anti-B	—
B	B	Anti-A	—
AB	A+B	None	Universal acceptor
O	None	Anti-A + Anti-B	Universal donor

RH SYSTEM & ERYTHROBLASTOSIS FETALIS

- **1939** Levine & Stetson demonstrated a new antibody in a transfusion-reactive, recently-delivered-stillborn woman's serum.

- **1940** Landsteiner & Wiener found rabbit anti-rhesus-monkey-RBC serum agglutinates 80% of normal human RBCs — named the **Rh (Rhesus) factor**.

Rh antigens determined by 3 closely linked allelomorphic gene pairs (Cc, Dd, Ee); **D (Rho)** is the most powerful, responsible for most Rh incompatibility. **Erythroblastosis fetalis**: Rh-negative dam carrying Rh-positive fetus develops antibody after transplacental leakage (minor in pregnancy, major at delivery); sensitisation occurs in first pregnancy; IgG anti-Rh antibodies cross the placenta in subsequent pregnancies, causing fetal haemolysis/jaundice.

MAJOR HISTOCOMPATIBILITY COMPLEX (MHC) **EXAM HOT**

Graft-antigen recognition causes rejection; potent histocompatibility antigens are inherited via the **MHC** gene cluster, controlling antigen processing/presentation. Genes labelled A, B, C, D — codominant; only one gene of each pair codes per individual.

SPECIES	MHC NAME
Human	HLA
Cattle	BOLA
Sheep	OLA
Horse	ELA
Swine	SLA
Dog	DLA
Mouse	H2
Chicken	B

CLASS I — nucleated cells, including B/T cells, macrophages, except erythrocytes

CLASS II — antigen-presenting cells: B lymphocytes, macrophages, dendritic cells

CLASS III — serum proteins, complement components

MHC restriction: T cells recognise foreign antigen only presented with self-MHC. T_c (CD8+) cells recognise antigen + MHC Class I on infected target cells; T_h (CD4+) cells recognise antigen + MHC Class II on APCs.

ANTIGEN PROCESSING & PRESENTATION

- **Exogenous** — macrophages/DCs/B cells phagocytose antigen, process in phagosome, fuse with MHC-II-containing endosome, present via MHC-II to CD4+ Th cells.
- **Endogenous** — viral/transformed-cell antigens processed by cytosolic immunoproteasomes; TAP1/TAP2 transport peptide to ER for MHC-I loading; presented on surface, recognised by Tc cells.
- **Non-peptide** — glycolipid/lipid antigens presented via CD1 family, non-classical Class-I-like molecules.



CHAPTER RECAP

- **Forssman antigen** — shared bacterial/RBC glycoprotein (horse, sheep, cat, mouse).
- **Weil-Felix reaction** — Proteus OX-19/OX-2/OX-K cross-reactive antigen basis.
- **Landsteiner, 1900** — discovered ABO blood groups.
- **O = universal donor, AB = universal acceptor.**
- **Rh D antigen** — most powerful, responsible for most Rh incompatibility.
- **MHC restriction** — CD8+/MHC-I vs CD4+/MHC-II pairing rule.
- **Species MHC names** — HLA (human), BOLA (cattle), SLA (swine), DLA (dog), H2 (mouse).

⚡ MASTER QUICK REVISION — 45 MUST-KNOW POINTS

1. **Jenner, 1798** — cowpox vaccination against smallpox; Father of Immunology.
2. **Pasteur, 1879-85** — attenuated vaccines: fowl cholera, anthrax, rabies.
3. **6 July 1885** — first human rabies vaccination (Meister); Zoonosis Day.
4. **Von Behring & Kitasato, 1890** — first proof of humoral immunity (antitoxin transfer).
5. **Landsteiner, 1901** — discovered ABO blood groups.
6. **Köhler & Milstein, 1975** — hybridoma/monoclonal antibody technology.
7. **Metchnikoff, 1882-83** — cellular theory; coined "phagocytosis".
8. **Buchner, 1889** — named heat-labile bactericidal factor "alexin" (complement).
9. **Thymus** — first lymphoid organ to develop; blood-thymus barrier in cortex.
10. **90-95% thymocyte apoptosis** — via positive/negative selection.

11. **Bursa of Fabricius** — bird-only B-cell primary organ; bursin hormone.
12. **Ileo-cecal Peyer's patches** — resemble avian bursa; gone by 15 months in sheep.
13. **Germinal centres** — B-cell somatic maturation, class switching, memory formation.
14. **HEV** — route for 90% of lymphocytes leaving a lymph node; absent in sheep.
15. **Thoracic duct** — largest lymph vessel; confirms T-cell recirculation on cannulation.
16. **PALS** — T-cell sheath around splenic central artery.
17. **Hemocatharesis** — red-pulp destruction of aged RBCs/platelets.
18. **Neutrophil granules** — primary (defensins, MPO) vs secondary (lysozyme, lactoferrin).
19. **Eosinophil ADCC** — IgE-Fc-mediated helminth killing.
20. **Mast cells vs basophils** — tissue-resident vs circulating, both IgE-bearing.
21. **Plasma cells** — secrete ~300 Ig molecules/second, live 2-3 days.
22. **γ/δ T cells** — ~60% of young ruminant blood lymphocytes.
23. **EAC rosette** — B-cell test; **E-rosette** — T-cell test (CD2).
24. **Respiratory burst** — NADPH oxidase → ROIs; myeloperoxidase forms hypohalides.
25. **RE system** — all mononuclear phagocytes (Kupffer, microglia, histiocytes etc).
26. **NK cells** — perforin + granzyme killing; activated when MHC-I is lost.
27. **Lymphocyte repertoire** — discriminates 10^7 - 10^9 antigenic determinants.
28. **Secondary response** — shorter lag, IgG-dominant vs IgM-dominant primary.
29. **Complement** — ≥ 19 proteins, heat-labile at $56^\circ\text{C}/30\text{ min}$.
30. **IgM > IgG** in complement fixation — one IgM molecule suffices.
31. **C4b2a** = classical C3 convertase; **C3bBb** = alternative C3 convertase.
32. **Properdin** — extends alternative C3 convertase half-life $5 \rightarrow 30\text{ min}$.
33. **MAC = C5b678(9)n** — transmembrane pore, osmotic lysis.
34. **Anaphylatoxins** — C3a, C4a, C5a trigger mast-cell degranulation.
35. **Immunogen $\subset$ Antigen** — all immunogens are antigens, not vice versa.
36. **9 antigenicity factors** — size, complexity, shape, stability, degradability, foreignness, genotype, dose/route, adjuvant.
37. **Freund's complete adjuvant** = incomplete + heat-killed Mycobacteria.
38. **Hapten** — antigenic only when carrier-conjugated.
39. **Epitope size** — 6-10 aa (T cell) vs 10-15 aa (B cell).
40. **Forssman antigen** — cross-reactive bacterial/RBC glycoprotein.
41. **Weil-Felix reaction** — Proteus OX antigen cross-reactivity with Rickettsiae.
42. **O = universal donor, AB = universal acceptor** (ABO system).
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